

Simultaneous Analog Placement and Routing with Current Flow and Current Density Considerations *

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ABSTRACT

Current-flow and current-density are two major considerations for placement and routing of analog layout synthesis. The current-flow constraints are specified to the critical nets with monotonic current/signal paths to reduce parasitic impacts. The current-density constraints are usually specified on the nets with variable wire widths to avoid the IR-drop and electromigration problems. In this paper, we propose the first work to simultaneously consider current-flow and current-density constraints while placing and routing the analog circuits with minimized chip area, routed wirelength, bend numbers, via counts, and coupling noise at the same time. We first present an enhanced **B*-tree representation** to simultaneously model modules and interconnects for an analog circuit. Then a simultaneous placement and routing algorithm is presented to generate a layout while satisfying the current-flow and current-density constraints with minimized chip area, routed wirelength, bend numbers, via counts, and coupling noise. Experimental results show that our approach can obtain better layout results and satisfy all specified constraints while optimizing circuit performance.

Categories and Subject Descriptors

B.7.2 [Integrated Circuits]: Design Aids [Placement and Routing]

General Terms

Algorithms, Performance

Keywords

Physical Design, Placement, Routing, Analog ICs

1. INTRODUCTION

In modern analog and mixed-signal circuit designs, *current-flow* and *current-density* are two major considerations for placement and routing of analog layout synthesis. The current-flow constraints are specified to the critical nets in the analog circuits with monotonic current/signal paths from the power line (VDD) to the ground line (GND) to increase the current smoothness and reduce the parasitic impacts on the current/signal paths. The current-density constraints are usually specified to the nets with variable wire widths according to the amount of current flowing the nets to avoid the IR-drop and electromigration

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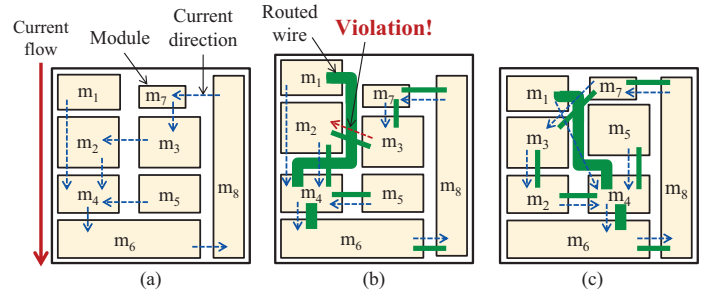


Figure 1: (a) A placement result satisfies the current-flow constraint. (b) The routing result of (a) with current-density consideration. (c) A simultaneous placement and routing result satisfies both current-flow and current-density constraints.

problems caused by high current densities. Moreover, the current-density constraints become more important to high-frequency analog circuits because the *skin effects* will confine the conducting region of a wire near the surface and further increase the current density.

Existing analog layout synthesis algorithms focus on one of the two constraints alone: current-flow constraints at the placement stage and current-density constraints at the routing stage. For the current-flow constraints, Long et al. proposed a signal-path driven partition and placement method for analog circuits [15], which simply places modules closely from left to right in each signal-path according to the pre-defined sequence to satisfy current-flow constraints. To achieve a more compact placement result, Wu et al. introduced a slicing-tree-based analog placement representation to generate multiple compact analog placements with current-path constraints [22]. For the current-density constraints, Alder et al. proposed a current driven router to calculate the required wire width of each signal net according to the current-density constraints on the fly, and route all signal nets with minimum detours in a single step [1, 2]. To further consider the electromigration problem, Lienig et al. introduced two two-stage global routing methodologies for current-driven routing to prevent electromigration in deep-sub-micron designs [10], and Lienig and Jerke further improved the routing efficiency by a current-dependent routing tree to determine the estimated routing paths [9].

To achieve better circuit performance and reduce the layout-induced parasitic impacts, it is desirable to simultaneously consider current-flow and current-density constraints at both of the placement and routing stages of analog layout synthesis. However, most of the traditional analog layout synthesis algorithms address on either placement or routing individually [3, 5, 6, 11, 13, 14, 16, 17, 18, 19, 21, 23, 24]. As a result, placement is performed followed by routing, leading to an ordered synthesis flow. This ordered analog layout synthesis flow cannot guarantee that a constraint (say, a current-flow constraint) addressed and satisfied at the placement stage will not be violated at the routing stage that considers another constraint (say, the current-density constraint). Figures 1(a) and (b) illustrate the placement and routing results considering current-flow and current-density constraints at the placement and routing stages in sequence. Since the interconnects usually cannot cross the active regions of devices [23], the current-flow con-

straints are violated after considering the current-density constraints at the routing stage as shown in Figure 1(b). The simultaneous placement and routing algorithm can simultaneously consider current-flow and current-density constraints and obtains a layout without any violation as shown in Figure 1(c).

Unfortunately, very few existing works, especially the topological-representation-based techniques, focus on simultaneous placement and routing of analog layout synthesis. Cohn et al. decomposed each interconnect into a two-terminal minimal-spanning tree, and incrementally reshaped or moved both modules and interconnects according to the change of the minimal-spanning trees during the annealing [7]. The later work [8] proposed a sequence-pair representation with a shaped-based formulation to simultaneously express modules and interconnects. Recently, Prieto et al. and Lin et al. worked on two-stage frameworks to consider the layout routability at the placement stage earlier [20, 12]. However, the incremental reshaping and moving may not be able to handle the multiple terminal nets, and also require some post-optimizations to further improve the routing quality. Although the sequence-pair with the shaped-based formulation can simultaneously perform placement and routing, it usually requires much time to obtain a layout and is often hard to extend the formulation to simultaneously consider current-flow and current-density constraints. Moreover, routability-driven analog placement usually uses a global router to evaluate its placement results, which might not be able to accurately capture the detailed routing results.

To simultaneously consider current-flow and current-density constraints in analog layout synthesis for better circuit performance, we propose an enhanced B*-tree representation to simultaneously model the modules and interconnects for an analog circuit while satisfying the current-flow and current-density constraints. According to the enhanced B*-tree representation, we also present a simultaneous placement and routing algorithm considering both current-flow and current-density constraints while placing and routing an analog circuit with minimized chip area, routed wirelength, bend numbers, via counts, and coupling noise at the same time.

The main contributions of this paper are summarized as follows:

- A new analog layout synthesis algorithm which simultaneously considers current-flow and current-density constraints is proposed. To the best of our knowledge, this is the first work that handles the two constraints simultaneously.
- A novel enhanced B*-tree representation for simultaneous placement and routing is presented. The enhanced B*-tree simultaneously models modules and interconnects in one representation while satisfying the current-flow and current-density constraints at the same time.
- A simultaneous placement and routing algorithm is proposed. The simultaneous placement and routing algorithm closes the gap between placement and routing and makes analog layout synthesis conform to the conventional design manner.
- Chip areas, routed wirelength, bend numbers, via counts, and coupling noise are simultaneously minimized in our simultaneous placement and routing algorithm.
- Experimental results show that our method can satisfy all the current-flow, current-density, and symmetry constraints, and achieve smaller chip area, shorter routed wirelength, and fewer vias than the previous work. The simulation results also show that our approach can achieve higher circuit performance than the previous work.

The remainder of this paper is organized as follows. Section 2 first reviews the existing hierarchical B*-tree representation and then formulates the analog layout synthesis problem with current-flow and current-density constraints. Section 3 introduces our proposed enhanced B*-tree representation. Section 4 presents our simultaneous analog placement and routing algorithm with current-flow and current-density considerations. Section 5 reports the experimental results. Finally, Section 6 concludes this paper.

2. PRELIMINARIES

In this section, we first review the hierarchical B*-tree [13] in Section 2.1, and then formally formulated the analog layout synthesis problem in Section 2.2.

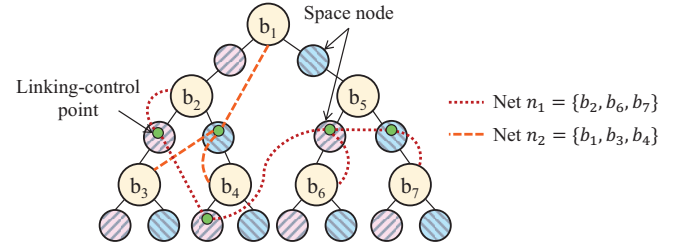


Figure 2: An example of the enhanced B*-tree representation.

2.1 Review of the Hierarchical B*-tree

The hierarchical B*-tree is an extension of B*-trees [4] that improving the ability of B*-trees to handle the symmetry constraints in analog placements. The hierarchical B*-tree is designed to represent a cluster of modules such as *symmetry islands*, *proximity groups*, or *pre-designed/existing circuits*. The function of a symmetry island and a proximity group is to connect the modules which are of the same symmetry group to each other to reduce the sensitivity to thermal gradients, process variations, or parasitic effects. The hierarchical B*-tree also inherits the benefits from the B*-tree, which requires only linear line to pack modules. The following subsection reviews the hierarchical nodes for symmetry islands and proximity groups.

2.1.1 Review of the Hierarchical Nodes for Symmetry Islands and Proximity Groups

To handle symmetry constraints, the hierarchical B*-tree introduces *hierarchical nodes* and uses them in a B*-tree to model an *automatically symmetric-feasible B*-tree* (ASF-B*-tree for short) for symmetry groups. Each ASF-B*-tree consists of sets of *symmetry modules* which are symmetrical with respect to the common symmetry axis or *self-symmetry modules* whose center lines are the symmetry axis. Since the symmetry constraints cannot be violated in a symmetry group, an ASF-B*-tree records only a half of the nodes in a symmetry group, and the relative positions of the remaining nodes can be calculated directly by mirroring the nodes in ASF-B*-tree. For the self-symmetry modules, their corresponding nodes in the ASF-B*-tree should always be on the right-most branch to avoid separating the two halves of the nodes from the symmetry axis.

2.2 Problem Formulation

We formally define the analog layout synthesis problem as follows:

- **The Analog Layout Synthesis Problem:** Given a set of modules, a netlist, a set of symmetry groups, a set of current-flow constraints, a set of current-density constraints, and the design rules, place all the modules and route all the nets to optimize the chip area, routed wirelength, bend number, via count, and coupling noise such that no design rule is violated and all the current-flow and current-density constraints are satisfied.

3. THE ENHANCED B*-TREE REPRESENTATION

To simultaneously consider current-flow and current-density constraints in analog layout synthesis, we propose a novel *enhanced B*-tree representation* to facilitate the simultaneous placement and routing while satisfying the current-flow and current-density constraints. Simultaneous placement and routing is not a trivial problem because the traditional analog placement representations might not be able to model modules and interconnects in one representation simultaneously. As a result, the traditional analog layout synthesis algorithms use the half-perimeter wirelength (HPWL) to estimate wirelength and routing congestions at the placement stage, and then apply a maze- or ILP-based router to connect all the interconnects while satisfying routing constraints. Although the feedback mechanism is applied from the router to the placer, it is still difficult for the placer to fulfill all the constraints from the feedback of the router since the interconnects usually cannot cross the active regions of devices. To remedy the deficiencies in traditional analog placement representations, the enhanced B*-tree representation is proposed to simultaneously model the modules and

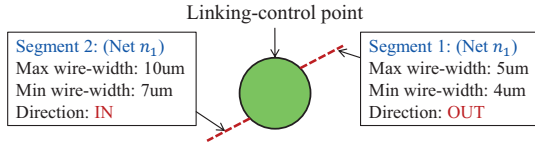


Figure 3: An example of a linking-control point. The information of each connected wire segment can be tracked in a linking-control point.

interconnects in one topological representation while facilitating simultaneous placement and routing and satisfying the current-flow and current-density constraints at the same time.

To facilitate the modelling of interconnects and the allocation of routing resources between modules, *linking-control points* and *space nodes* are introduced in our enhanced B*-tree representation. Figure 2 shows an example of the enhanced B*-tree representation. The linking-control points are used to link the interconnects and control the routing topologies between modules, and space nodes are used to reserve and allocate routing resources for interconnects. The linking-control points appear in the middle of any interconnects and can only be placed inside the space node, and the space nodes appear on the left and right children of each tree node b_i . The space node and the linking-control point will be detailed in the following subsections.

3.1 Linking-Control Point

To control routing topologies and reserve routing resources between modules in a topological representation, the linking-control point is introduced in our enhanced B*-tree representation. The concept of the linking-control point in the enhanced B*-tree is similar to the *Steiner point* in the geometric routing space, which means that the routing topologies in an enhanced B*-tree can be changed by moving or inserting linking-control points to different space nodes to achieve better placement and routing results. A linking-control point can only connect the wire segments belonging to the same interconnect, and each linking-control point has at least two wire segments. When a linking-control point connects only two wire segments, the linking-control point can be regarded as a *physical bending point*, which can increase the degree of freedom for interconnects. Figure 3 shows an example of a linking-control point. The linking-control point also contains the information of each connected wire segment, such as the minimum/maximaum wire width for current-density constraints and the current direction for current-flow constraints. This information provides an important basis for resource allocation, constraint satisfaction, and circuit optimization in the enhanced B*-tree representation.

3.2 Space Node

To reduce routing congestion and satisfy the current-flow and current-density constraints, routing resource reservation and allocation play an important role in the simultaneous placement and routing. Since the interconnects usually cannot cross the active regions of devices, the routing resource is very limited in an analog circuit. Thus, we introduce the *space node* for each tree node of the enhanced B*-tree to reserve and allocate routing resource for interconnects efficiently.

Figure 4(a) illustrates the corresponding space nodes of the tree node b_1 . For each tree node in the enhanced B*-tree, there are two space nodes on the left child (left space node for short) and the right child (right space node for short). The left and right space nodes correspond to the right and top space besides the module (Figure 4(b)), respectively. Different from the previous work [12], our space node employs linking-control points to provide a substantial basis for resource reservation and allocation, which can effectively reflect the required routing resource to route the interconnects with variable wire widths and current directions, and guide the enhanced B*-tree to a better layout result.

In addition, each right or top space contains several *physical location candidates* for a linking-control point, (inside the space node) to represent its *physical location* in an enhanced B*-tree as shown in Figure 4(c). The physical location candidates can be defined according to the number of linking-control points, the required wire width, and pin locations, or can be accurately defined by analog designers for some specific patterns. With linking-control points and space nodes, the

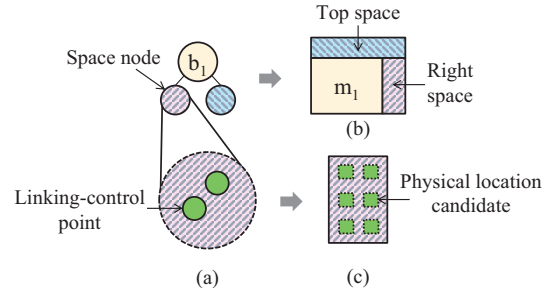


Figure 4: (a) The tree node b_1 has two space nodes on its left and right children, and each space node might contain several linking-control points. (b) The right and top space beside the module m_1 . (c) The corresponding physical location candidates on the right space for the linking-control points.

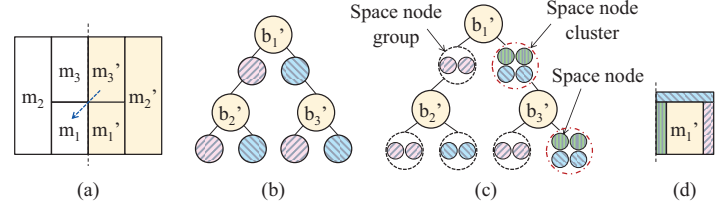


Figure 5: (a) A symmetry island with one interconnect. The corresponding symmetry net is ignored in this example. (b) The corresponding ASF-B*-tree with space nodes for the left symmetry island. (c) An enhanced ASF-B*-tree with both space node group and space node cluster for the left symmetry island. (d) The module m_1' with the corresponding space node cluster.

analog layout synthesis problem can be easily solved by our enhanced B*-tree representation effectively and efficiently. The routing resource allocation and the physical location determination will be introduced in Section 4.2 and Section 4.3, respectively.

3.3 Symmetry Constraint Handling

As mentioned in Section 2.1.1, the symmetry modules can be modelled by an ASF-B*-tree to satisfy the symmetry constraints; however, the ASF-B*-tree records only a half of the nodes in a symmetry group. As a result, the space node proposed in Section 3.2 cannot be applied in the ASF-B*-tree directly since the corresponding left/top space of the other half nodes cannot be represented in the ASF-B*-tree. Figures 5(a) and (b) show an example of a symmetry-island and the corresponding ASF-B*-tree. For the interconnect in Figure 5(a), the linking-control points of that interconnect cannot explore the spaces located on the top and left of the module m_1 in the ASF-B*-tree (Figure 5(b)).

To remedy the deficiencies, we introduce *space node groups* and *space node clusters* to improve the ASF-B*-tree. Figure 5(c) shows an example of the enhanced ASF-B*-tree. The space node group in the enhanced ASF-B*-tree is formed by two space nodes to represent the right (left) and top spaces for the symmetry modules in both sides. The space node cluster has four space nodes to represent not only the right (left) and top spaces for the symmetry modules in both sides but also the space between modules, which abut on a symmetry axis (Figure 5(d)). According to the properties of the ASF-B*-tree, the space node cluster exists only on the right-most branch for the modules abutting on a symmetry axis.

4. SIMULTANEOUS PLACEMENT AND ROUTING

In this section, we propose a simultaneous placement and routing algorithm based on the enhanced B*-tree representation to si-

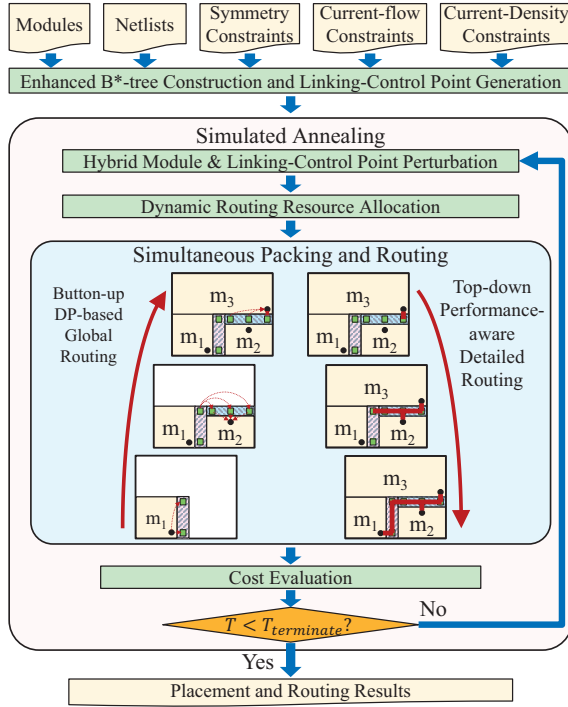


Figure 6: The simultaneous placement and routing algorithm.

multaneously consider current-flow and current-density constraints. Figure 6 shows our proposed simultaneous placement and routing flow. After acquiring the input modules, netlist, and current-flow and current-density constraints, we first construct an enhanced B*-tree and the corresponding linking-control points. The symmetry groups, pre-designed/existing circuits, and user-specified design hierarchies are modelled as corresponding enhanced ASF-B*-trees based on the statistics of symmetry and design hierarchy constraints. This step is followed by the simulated annealing scheme to seek for a desired layout. To facilitate the simultaneous placement and routing at the tree perturbing stage, hybrid perturbations are also proposed. Then, dynamic routing resource allocation is performed to allocate suitable routing space and locations for each interconnect according to the enhanced B*-tree. After the dynamic routing resource allocation, simultaneous packing and routing is presented with a bottom-up dynamic-programming-based global routing and a top-down performance-aware detailed routing to obtain the best physical locations of modules and linking-control points while optimizing routing results and satisfying the current-flow and current-density constraints. In addition to the chip area, our simultaneous placement and routing algorithm also optimized the routed wirelength, bend number, via count, and coupling noise at the same time.

4.1 Hybrid Perturbations

To perturb both modules and interconnects in the enhanced B*-tree, the hybrid perturbations for both modules and linking-control points are proposed. For the original three perturbations of the B*-tree, *delete* and *insert* (delete a node and insert it to a different position), *swap* (exchange the positions of two nodes), and *rotate* (rotate a node by 90 degrees clockwise), the space nodes and the linking control points, which inside the space nodes, are regarded as a part of the tree node, and must be moved with the tree node together. Figure 7(a) and (b) show an example of the delete and insert operation for the enhanced B*-tree.

In addition to the original three perturbations, we also propose four new moves for the linking-control points to perturb the routing topologies simultaneously as follows:

- *Delete and insert*. Delete a linking-control point and insert it to another position. Figures 7(b) and (c) show an example of the change before and after the delete and insert operation.

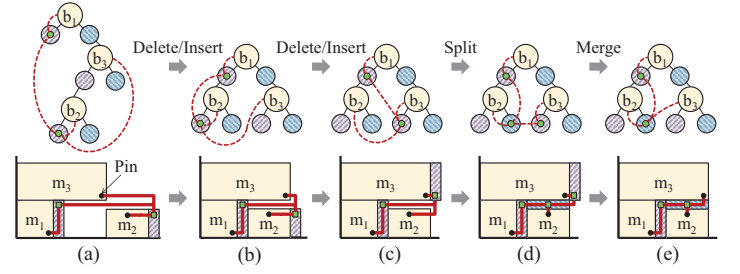


Figure 7: The enhanced B*-tree with the corresponding layout. (a) Before deleting/inserting b_2 . (b) After deleting/inserting b_2 . (c) After deleting/inserting the linking-control point. (d) After splitting the linking-control point. (e) After merging the linking-control points.

- *Swap*. Exchange the positions of two linking-control points.
- *Split*. Split one linking-control point into two linking-control points. The wire segments connecting to the original linking-control point will also be distributed to the two linking-control points. Figures 7(c) and (d) show an example of the change before and after the split operation.
- *Merge*. Combine two linking-control points into a new linking-control point. The wire segments belonging to the original two linking-control points will also be combined and connected to the new linking-control point. Figures 7(d) and (e) show an example of the merge operation.

4.2 Dynamic Routing Resource Reservation and Allocation

To acquire suitable routing resource and compensate the routing resource blocked by existing circuits or active regions, we propose a dynamic routing resource reservation and allocation technique to reserve and allocate appropriate routing spaces and locations according to space nodes and linking-control points after the hybrid perturbations. As mentioned in Section 3.1, the required maximum wire widths and current directions by each linking-control point can be calculated by the information of each wire segment recorded in the linking-control point. Therefore, the required total routing space by a space node can be dynamically reserved and allocated after the hybrid perturbations. Assume that R_i is the routing space required by the space node i , L_j is the maximum wire width required by the linking-control point j , and K_j is the number of wire segments connecting to the linking-control point j . For each space node i , the required routing space R_i can be defined as follows:

$$R_i = \sum_{j \in i} \left(\frac{L_j \times K_j}{2} \right). \quad (1)$$

In addition to the summation of the maximum wire width required by each linking-control point, the coupling noise induced by congested wire segments should also be considered. For this reason, we add the number of connected wire segments in Equation (1) to allocate more routing space to a space node. Since each linking-control point connects at least two wire segments, we divide the number of connected wire segments by two to avoid overestimation.

4.3 Simultaneous Packing and Routing

To place and route analog circuits with the enhanced B*-tree representation simultaneously, we propose a simultaneous packing and routing technique to generate placement and routing results at the same time. Different from the previous ordered analog layout synthesis algorithms (placement followed by routing), our enhanced B*-tree with the proposed simultaneous packing and routing technique requires only one stage to simultaneously place and route an analog circuit while optimizing its chip area, routed wirelength, bend number, via count, and coupling noise.

The simultaneous packing and routing technique consists of two phases, the bottom-up dynamic-programming-based (DP-based for

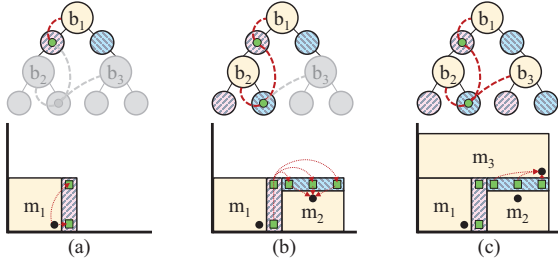


Figure 8: An example of the DP-based global routing. All the possible routing topologies and costs between two terminals are recorded by the dynamic programming as shown in the red-dotted arrows. (a) Packing the first module m_1 . (b) Packing the second module m_2 . (c) Packing the third module m_3 .

short) global routing and the top-down performance-aware detailed routing. The bottom-up DP-based global routing is performed with an integrated packing procedure to place the modules while determining the best physical location for each linking-control point beside a module. After packing the modules, the performance-aware detailed routing is performed with the back trace of the dynamic programming to connect all the wire segments while minimizing the routed wirelength, bend number, via count, and coupling noise.

4.3.1 Dynamic-Programming-Based Global Routing

To reduce constraint violations caused by linking-control points, we propose a bottom-up dynamic-programming-based (DP-based) global routing method, which integrates the packing procedure from B*-trees, to place modules while globally exploring the best physical locations for linking-control points and satisfying the current-flow and current-density constraints.

Figure 8 illustrates a DP-based global routing procedure. Starting from the root node b_1 , the coordinate of each module and its space node can be calculated by traversing a given enhanced B*-tree in a pre-order style. To determine the best physical location of a linking-control point, the dynamic programming is applied to construct a sub-problem which records all possible routing topologies with respect to all the physical location candidates of the linking-control point. As the red-dotted arrows shown in Figure 8, when the two terminals of a wire segment are connected by modules or linking-control points during packing, the dynamic programming will be activated to record all possible routing topologies and the corresponding costs between these two terminals. The current-flow constraints can be checked by examining the current directions recorded by the linking-control point in constant time during the DP-based global routing. Assume that C_{ij} is the cost to connect the wire segment i with a possible routing topology j , W_{ij} and B_{ij} are the estimated wirelength and bend numbers to connect the wire segment i with the routing topology j by A* search, and p is a penalty for unsuccessful routing or constraint violations, the cost C_{ij} can be defined as follows:

$$C_{ij} = \alpha W_{ij} + \beta B_{ij} + p. \quad (2)$$

If the A* search fails to find a feasible path or the path violates the current-flow constraint, a high penalty will be given to avoid the routing topology to be chosen by the dynamic programming.

4.3.2 Performance-Aware Detailed Routing

For better circuit performance, we propose a top-down performance-aware detailed routing after the bottom-up DP-based global routing to simultaneously trace back the sub-problems constructed by dynamic programming and detailed route the chosen wire segments. Since a routing topology might have branches at the linking-control points, we trace back the sub-problems from a leaf of an enhanced B*-tree, and follow the reverse packing order to avoid rearranging the objects. To honor the symmetry constraints and expert knowledge, the symmetry and critical nets will be traced back and routed first. During the back trace, the sub-problem which has the lowest cost and the minimum coupling effect and satisfies the current-flow and current-density constraints will be chosen and simultaneously routed.

To obtain a better layout by our enhanced B*-tree, we integrate the routing cost with the chip area to guide the simulated annealing scheme to explore more practical solutions. The total cost ϕ for the enhanced B*-tree can be defined as follows:

$$\phi = \alpha \frac{A}{\hat{A}} + \beta \frac{W}{\hat{W}} + \gamma \frac{B}{\hat{B}} + \delta \frac{V}{\hat{V}} + p, \quad (3)$$

where $\alpha, \beta, \gamma, \delta$ are user-specified weights, $A, W, B, V, \hat{A}, \hat{W}, \hat{B}, \hat{V}$ are the chip area, routed wirelength, bend number, via count, normalized chip area, normalized routed wirelength, normalized bend number, and normalized via count, respectively. p is a penalty for unsuccessful routing or constraint violations.

Table 1: Design statistics and constraint number of Comparator and Two-stage OP.

Circuits	Module	Symmetry	Current-Flow	Current-Density
Comparator	12	2	6	14
Two-stage OP	14	3	4	12

5. EXPERIMENTAL RESULTS

We implemented our simultaneous placement and routing algorithm in the C++ programming language, and implemented the algorithm in [22] in the C programming language. All experiments were performed on an Intel Xeon E5-2620 2.0GHz Linux workstation with 72GB memory. To evaluate our approach, we used two analog circuits, Comparator and Two-stage OP. These two circuits have been tape-out by the TSMC 90nm MSG process manually by expert analog designers. The statistics of the two circuits are shown in Table 1. We also applied the TSMC 90nm MSG library and used two layers to route the two circuits. After generating the layout automatically by our simultaneous placement and routing algorithm, we used Calibre nmDRC and Calibre nmLVS to check the design rules and verify the layout versus schematic.

Since there is no existing work simultaneously handling the current-flow and current-density constraints for simultaneous placement and routing, we compared our work with [22] for chip area, routed wirelength, and via count minimization. Since the current-density constraint and the routing problem are not considered in [22], we asked an expert analog designer to manually route the placement result generated by the work [22], and slightly moved the placed modules to allocate suitable area for routing without violating the current-flow/current-path and current-density constraints. As shown in Table 2, our analog layout synthesis algorithm can obtain an average chip area reduction of 7%, an average routed wirelength reduction of 12%, and an average via reduction of 8%. These results show that our enhanced B*-tree, which simultaneously models modules and interconnects, can facilitate the simultaneous placement and routing algorithm to further reduce the chip area, routed wirelength, and via count while satisfying the current-flow and current-density constraints effectively. Since there are some current-density constraints with variable wire widths in both circuits, the modules in the placement result of [22] must be moved to allocate suitable routing area, resulting in a larger chip area and longer routed wirelength. The running time of our approach is also reported in Table 2. Note that we do not compare the running time with the work [22] since the work handles only the placement problem.

We also ran post-layout simulation to verify the quality of our automatic layout designs. We obtained the netlist with extracted parasitic RC by using Calibre xRC in the post-layout simulation, and simulated the netlist with extracted parasitic RC by using HSPICE after the RC extraction. The rise time delay, fall time delay, propagation delay, DC gain, unity gain bandwidth, and phase margin were also measured by HSPICE. Tables 3 and 4 show the simulation results of Comparator and Two-stage OP, respectively. In both results, our approach can satisfy all specified current-flow, current-density, and symmetry constraints. Since the work [22] handles only the current-flow/current-path and symmetry constraints at the placement stage, it is predictable that our approach simultaneously considering current-flow, current-density, and symmetry constraints results in much more

Table 2: Comparison of the layout results between the work [22] and our approach.

Circuits	[22]			Ours			
	Area (μm^2)	Wirelength (μm)	Via Number	Area (μm^2)	Wirelength (μm)	Via Number	Runtime (s)
Comparator	285.7	95.6	59	277.1	82.3	51	242.8
Two-stage OP	695.1	148.3	31	632.5	136.5	31	255.7
Comp.	1.07	1.12	1.08	1.00	1.00	1.00	

Table 3: Comparison of the rise time delay, fall time delay, average delay, and propagation delay between [22] and our approach.

Circuit	t_{dr} (ns)		t_{df} (ns)		t_{davg} (ns)		t_{plh} (ns)		t_{phl} (ns)	
	[22]	Ours	[22]	Ours	[22]	Ours	[22]	Ours	[22]	Ours
Comparator	1.23	1.21	1.85	1.82	1.54	1.52	0.68	0.67	0.97	0.96

Table 4: Comparison of the DC Gain, Unity Gain Bandwidth, and Phase Margin between [22] and our approach.

Circuits	DC Gain (dB)		Unity Gain Bandwidth (MHz)		Phase Margin ($^\circ$)	
	[22]	Ours	[22]	Ours	[22]	Ours
Two-stage OP	40.28	40.3	17.6	17.62	53.54	53.48

compact layouts, and further improves the current smoothness and reduces the parasitic impacts and IR drops on the interconnects. Moreover, our work can reduce the rise time delay, fall time delay, average delay, and propagation delay over the work [22] for Comparator. Besides, our work can also improve the DC gain and unity gain bandwidth over the work [22] for Two-stage OP. In addition, our phase margin is closest to the specification (45°). Figure 9 shows the resulting layout of Comparator generated by our approach.

6. CONCLUSIONS

In this paper, we have proposed a simultaneous placement and routing algorithm with current-flow and current-density constraints for analog circuit designs. To facilitate the simultaneous placement and routing, the enhanced B*-tree has also been proposed. Our algorithm not only minimizes the chip area but also reduces the routed wirelength, bend number, via count, and coupling noise at the same time. Experimental results have shown that our proposed algorithm is effective and efficient for analog layout synthesis.

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8. REFERENCES

- [1] T. Adler, and E. Barke, "Single Step Current Driven Routing of Multiterminal Signal Nets for Analog Applications," *Proc. of DATE*, Mar. 2000.
- [2] T. Adler, H. Brocke, L. Hedrich, and F. Barke, "A Current Driven Routing and Verification Methodology for Analog Applications," *Proc. of DAC*, Jun. 2000.
- [3] F. Balasa and K. Lampaert, "Symmetry Within the Sequence-Pair Representation in the Context of Placement for Analog Design," *IEEE TCAD*, vol. 19, no. 7, Jul. 2000.
- [4] Y.-C. Chang, Y.-W. Chang, G.-M. Wu, and S.-W. Wu, "B*-Trees: A New Representation for Non-slicing Floorplan," *Proc. of DAC*, Jun. 2000.
- [5] P.-Y. Chou, H.-C. Ou, and Y.-W. Chang, "Heterogeneous B*-trees for Analog Placement with Symmetry and Regularity Considerations," *Proc. of ICCAD*, Nov. 2011.
- [6] J. M. Cohn, D. J. Garrod, R. A. Rutenbar, and L. R. Carley, "KOAN/ANAGRAM II: New Tools for Device-level Analog Placement and Routing," *IEEE JSSC*, vol. 26, pp. 330–342, Mar. 1991.
- [7] J. M. Cohn, D. J. Garrod, R. A. Rutenbar, and L. R. Carley, "Techniques for Simultaneous Placement and Routing of Custom Analog Cells in KOAN/ANAGRAM II," *Proc. of ICCAD*, Nov. 1991.
- [8] Y. Kubo, S. Nakatake, Y. Kajitani, and M. Kawakita, "Explicit Expression and Simultaneous Optimization of Placement and Routing for Analog IC Layouts," *Proc. of VLSID*, Jan. 2002.
- [9] J. Lienig, and G. Jerke, "Current-driven Wire Planning for Electromigration Avoidance in Analog Circuits," *Proc. of ASP-DAC*, Jan. 2003.
- [10] J. Lienig, G. Jerke, and T. Adler, "Electromigration Avoidance in Analog Circuits: Two Methodologies for Current-driven Routing," *Proc. of ASP-DAC*, Jan. 2002.
- [11] C.-W. Lin, J.-M. Lin, C.-P. Huang, and S.-J. Chang, "Performance-driven Analog Placement Considering Boundary Constraint," *Proc. of DAC*, Jun. 2010.
- [12] C.-W. Lin, C.-C. Lu, J.-M. Lin, and S.-J. Chang, "Routability-driven Placement Algorithm for Analog Integrated Circuits," *Proc. of ISPD*, Mar. 2012.
- [13] P.-H. Lin, Y.-W. Chang, and S.-C. Lin, "Analog Placement Based on Symmetry-Island Formulation," *IEEE TCAD*, vol. 28, no. 6, pp. 791–804, Jun. 2009.
- [14] M. P.-H. Lin, H. Zhang, M. D. F. Wong, and Y.-W. Chang, "Thermal-Driven Analog Placement Considering Device Matching," *IEEE TCAD*, vol. 30, no. 3, pp. 325–336, Mar. 2011.
- [15] D. Long, X. Hong, and S. Dong, "Signal-Path Driven Partition and Placement for Analog Circuit," *Proc. of ASP-DAC*, Jan. 2006.
- [16] Q. Ma, L. F. Xiao, Y.-C. Tam, and Evangeline F. Y. Young, "Simultaneous Handling of Symmetry, Common Centroid, and General Placement Constraints," *IEEE TCAD*, vol. 30, no. 1, pp. 85–95, Jan. 2011.
- [17] E. Malavasi, E. Charbon, E. Felt, and A. Sangiovanni-Vincentelli, "Automation of IC Layout with Analog Constraints," *IEEE TCAD*, vol. 15, no. 8, pp. 923–942, Aug. 1996.
- [18] H.-C. Ou, H.-C. Chang Chien, and Y.-W. Chang, "Non-Uniform Multilevel Analog Routing with Matching Constraints," *Proc. of DAC*, Jun. 2012.
- [19] M. M. Ozdal and R. F. Hentschke, "Maze Routing Algorithms with Exact Matching Constraints for Analog and Mixed Signal Design," *Proc. of ICCAD*, Nov. 2012.
- [20] J. A. Prieto, A. Rueda, J. M. Quintana and J. L. Huertas, "A Performance-Driven Placement Algorithm with Simultaneous Place&Route Optimization for Analog IC's," *Proc. of EDTC*, Mar. 1997.
- [21] H.-F. Tsao, P.-Y. Chou, S.-L. Huang, Y.-W. Chang, Mark P.-H. Lin, D.-P. Chen, and D. Liu, "A Corner Stitching Compliant B*-tree Representation and Its Applications to Analog Placement," *Proc. of ICCAD*, Nov. 2011.
- [22] P.-H. Wu, Mark P.-H. Lin, Y.-R. Chen, B.-S. Chou, T.-C. Chen, T.-Y. Ho, and B.-D. Liu, "Performance-driven Analog Placement Considering Monotonic Current Paths," *Proc. of ICCAD*, Nov. 2012.
- [23] L. F. Xiao, Evangeline F. Y. Young, X.-Y. He, and K.P. Pun, "Practical Placement and Routing Techniques for Analog Circuit Designs," *Proc. of ICCAD*, Nov. 2010.
- [24] L. Zhang, U. Kleine, and Y. Jiang, "An Automated Design Tool for Analog Layouts," *IEEE TVLSI*, vol. 14, no. 8, pp. 881–894, Aug. 2006.

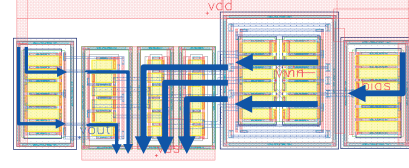


Figure 9: The layout of Comparator generated by our approach.